🧑‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 10 · STANDARD 6.GR.2

Volume with Whole Number Edges Flagship

Lesson 10-1-flagship · Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period

CLASS LEGACY MISSION

Seal the Time Capsule

You are the lead builder for the school's 50-year time capsule project. Every memory must fit inside rectangular containers, and the principal needs to know exactly how much each one holds before they are buried. Master volume and the capsule gets sealed for the future.

My Notes

Level 3 Enrichment

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can find the volume of a rectangular prism with whole-number edges using length × width × height.

Language Objective: I can explain my work using the words volume, rectangular prism, cubic units, and edge.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Volume | Rectangular prism | Cubic units | Length, width, height | Net | Edge

👆 *Tap any word to see what it means and a picture.*

Level 3: try the blanks from memory first, then check the bank.

1. The amount of space inside a three-dimensional solid is its ________.

2. A solid with six rectangular faces, like a box, is a ________.

3. The unit used to measure volume, like cubic centimeters, is ________.

4. The three edge measurements of a rectangular prism are its ________.

5. A flat pattern that folds up into a solid is a ________.

6 A line segment where two faces of a solid meet is an ______________.

Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

1. Understand the Question

EXPLAIN

In the Fold Shop you fold a flat net into a closed box. How does folding the net help you understand what the volume $V = l \times w \times h$ actually measures?

Your job: Answer the question. Use math evidence. Explain how the evidence proves your answer.

Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.

2. Plan Your Math Words

Check at least two words you will use.

- [ ] **Volume** — How much space is inside a solid shape.
  Volumen — Cuánto espacio hay dentro de una figura sólida.
- [ ] **Rectangular prism** — A solid box shape with six flat rectangle sides.
  Prisma rectangular — Una figura sólida en forma de caja con seis lados rectangulares.
- [ ] **Cubic units** — The units used to measure space inside, like cubic inches.
  Unidades cúbicas — Las unidades para medir el espacio interior, como pulgadas cúbicas.
- [ ] **Net** — A flat shape that folds up into a solid.
  Plantilla (desarrollo plano) — Una figura plana que se dobla y forma un sólido.
- [ ] **Edge** — The line where two flat sides of a solid meet.
  Arista — La línea donde se unen dos lados planos de un sólido.

Say it first: Say your idea to a partner or quietly to yourself before you write.
Di tu idea a un compañero o en voz baja antes de escribir.

When the net folds up, the ____ faces enclose a ____ that I can fill with unit cubes.
Cuando la red se dobla, las ____ caras encierran un ____ que puedo llenar con cubos unitarios.

3. Build Your Explanation

Model the parts: This model shows the parts of an explanation. It does not answer your writing question.

Claim: Volume measures space in three directions, so you need length, width, and height.

Evidence: Students explain volume is three-dimensional, so all three edges (length, width, height) are needed; two measurements only describe one flat face.

Reasoning: This evidence connects the definition of volume to the mathematical claim.

Start

Most language support

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **When the net folds up, the ____ faces enclose a ____ that I can fill with unit cubes.**
  Cuando la red se dobla, las ____ caras encierran un ____ que puedo llenar con cubos unitarios.
- **The volume is the number of ____ that fit inside, found by ____ × ____ × ____.**
  El volumen es el número de ____ que caben dentro, hallado por ____ × ____ × ____.

Build

Some language support

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**
  Mi afirmación es ____.
- **I know because ____.**
  Lo sé porque ____.
- **So ____.**
  Entonces ____.

Explain

Light language support

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- **My claim is ____.**
  Mi afirmación es ____.
- **My math evidence is ____.**
  Mi evidencia matemática es ____.
- **This proves ____ because ____.**
  Esto demuestra ____ porque ____.

4. Check Your Explanation

- [ ] I answered the question.
  Respondí la pregunta.
- [ ] I used math evidence: numbers, an equation, a model, or a comparison.
  Usé evidencia matemática: números, una ecuación, un modelo o una comparación.
- [ ] I explained how the evidence proves my answer.
  Expliqué cómo la evidencia demuestra mi respuesta.
- [ ] I used at least two math words.
  Usé por lo menos dos palabras matemáticas.
- [ ] I reread complete sentences and punctuation.
  Volví a leer mis oraciones completas y la puntuación.

Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

Answer Key & Teacher Guide

1. **Notes 1:** Volume
2. **Notes 2:** Rectangular prism
3. **Notes 3:** Cubic units
4. **Notes 4:** Length, width, height
5. **Notes 5:** Net
6. **Notes 6:** Edge
7. **Watch for this mistake:** *The most common mistake in Volume with Whole Number Edges is adding the three edges instead of multiplying them — writing* $6 + 4 + 3 = 13$ *instead of* $6 \times 4 \times 3 = 72$*. Volume counts the unit cubes that pack a 3D space, so all three edges must be multiplied, not added, and the answer must be labeled in cubic units (like* in^3*), not square units or a bare number.*
8. **Try It 1:** A. 3 in — $V = l \times w \times h \rightarrow 72 = 6 \times 4 \times h \rightarrow 72 = 24 \times h \rightarrow h = 72 \div 24 = 3$ *in.*
9. **Try It 2:** A. 200 cubic inches — $V = l \times w \times h = 10 \times 4 \times 5 = 200$ *cubic inches. Volume always uses cubic units because it fills space in three directions.*